INCH-POUND
MIL-PRF-3098/131D
27 August 1997
SUPERSEDING
MIL-C-3098/131C
24 September 1992

PERFORMANCE SPECIFICATION SHEET

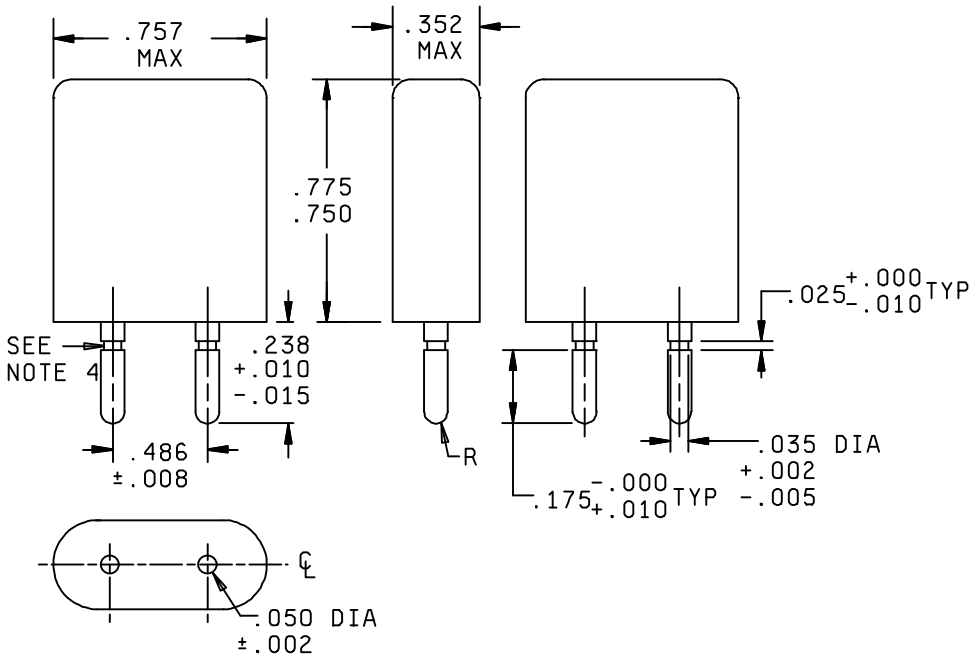
CRYSTAL UNIT, QUARTZ, CR150/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 45 MHz to 100 MHz; fifth mechanical overtone; controlled; series resonance.

Inches	mm
.002	.05
.005	.13
.008	.20
.010	.25
.025	.64
.035	.89
.050	1.27
.238	6.05
.352	8.94
.486	12.34
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR150/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 45.0 MHz to 100 MHz, inclusive.

Operable temperature range: +55°C to +70°C, inclusive.

Operating temperature range (controlled): +70°C to +80°C, inclusive.

Frequency and equivalent resistance:

Equivalent resistance: 40 ohms, maximum.

Reference temperature: 75°C ±1°C.

Frequency tolerance: ±5 ppm.

Mode of oscillator: Fifth mechanical overtone.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 3.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

Thermal shock:

Frequency change permitted: ±2.5 ppm.

Equivalent resistance change permitted: ±40 ohms, maximum.

NOTE: CF-150/U is similar to CR-143/U except for a wider frequency tolerance.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC

Review activities:
Army - AR, MI

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0697-85)

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